



Research Paper

Comparison of Adsorption Capacity of Natural and Acid-activated Kaolinite Clay for Cesium ions from Aqueous Solution

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Objectives : This study is to compare the removal efficiency of cesium ions in aqueous solutions by natural and acid-activated kaolinite clay.

Methods : Natural kaolinite clay was acid-treated with H_2SO_4 (2 M) at 80°C for 6 h under mechanical stirring. While activating natural kaolinite clay with acid, cations such as Al^{3+} , Ca^{2+} , Mg^{2+} and Fe^{3+} were partially eluted from the crystal lattice of natural kaolinite clay, and resulted in the increase in the surface area and the pore volume through the opening of crystal lattice. The surface area and the pore volume of acid-activated kaolinite clay were found to be roughly three times higher than natural kaolinite clay. The characteristics of natural and acid-activated kaolinite clay were observed by X-ray Fluorescence Spectroscopy, Energy Dispersive X-ray Spectroscopy, and BET Surface Area Analyzer.

Results and Discussion : Generally, adsorption efficiency of Cs^+ ion by acid-activated kaolinite clay showed much higher compared with natural kaolinite clay. At 50 mg L^{-1} of Cs^+ ion concentration and the unit of dose in g L^{-1} , the adsorption efficiencies of Cs^+ ion by natural and acid-activated kaolinite clay were 57.5% and 96.9%, respectively. The data obtained from this study were fitted to the adsorption isotherm and the kinetic models, respectively. It revealed that the Langmuir isotherm and the pseudo-second-order kinetic models described well the adsorption behavior of Cs^+ ions on both natural and acid-activated kaolinite clay owing to their higher correlation coefficient R^2 . Based on the Langmuir isotherm coefficient Q , adsorption capacity of Cs^+ ion by natural kaolinite clay and acid-activated kaolinite clay were 5.65 mg g^{-1} and 10.6 mg g^{-1} , respectively.

Conclusion : The results demonstrated that acid-activated kaolinite clay with acid treatment can be used as more effective adsorbent for the adsorption of Cs^+ ions from aqueous solution than natural kaolinite clay.

Keywords : Acid-activated kaolinite clay, Adsorption, Cs^+ ions, Isotherm, kinetic

The Korean text of this paper can be translated into multiple languages on the website of <http://jksee.or.kr> through Google Translator.

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1. Introduction

The radioactive ^{137}Cs has been a major concern due to its high radioactivity and long half-life (30.2 years).¹⁾ In 2011, the disaster of Fukushima Daiichi Nuclear Power Plant occurred in the aftermath of the earthquake and tsunami in Japan, resulting in a huge amount of Cs released into the surrounding environment. Because of this accident, the total amount of Cs emitted during the Fukushima nuclear accident was estimated to be from 15 to 30 PBq.²⁾ Among isotopes of Cs, ^{137}Cs has been used as an indicator of radioactive contamination because it hardly exists in nature and is mostly generated from nuclear power generation or the use of nuclear weapons.³⁾ $^{137}\text{Cs}^+$ ions exhibit behavior similar to potassium ions (K^+) in the body, so when absorbed into the body, it is not easily discharged and is highly concentrated on the organism through several stages of the food chain, causing various diseases including infertility, paralysis, lung and bone marrow cancers.⁴⁾ A number of techniques such as ion exchange, liquid extraction, membrane separation, chemical precipitation, coagulation, reverse osmosis filtration, adsorption, and electrochemical separation have been proposed for removing Cs^+ ion in liquid radioactive waste.^{5,6)} Among the techniques for removing Cs^+ ions from liquid radioactive waste, adsorption has proven to be economical and convenient to operate as well as the usefulness of a wide range of adsorbents.⁷⁾ Materials such as polymers, biomass, and metal oxides as adsorbents have been evaluated for the removal of Cs^+ ions from liquid radioactive waste.⁸⁻¹²⁾ However, although these adsorbents are known to be very effective in removing Cs^+ ions, they have disadvantages such as cost and process flexibility for manufacturing the adsorbent.

Over several decades, special attention has been continuously focused on the use of natural resources as replacements for conventional adsorbents. Among the alternative adsorbents, clay stands out in particular because of its abundance in surrounding area and the lower cost compared to activated carbon, zeolites, and synthetic exchange resins.^{13,14)} It has been reported that clay materials have been used for the removal of various metals such as Cu^{2+} , Pb^{2+} , Cd^{2+} , Zn^{2+} , Co^{2+} from aqueous solution by providing a large amount of cation exchange capacity in its layer structure.^{15,16)} However, in general, natural clay itself showed relatively low efficiency in removing such metals ion from aqueous solution. Therefore, there is a need for a process to improve the adsorption capacity of natural clay through activation. One of the most appropriate

activation methods is an acid treatment process.^{17,18)} Acid-activated clay has been used in various industries for pollutant removal due to their wide range of outstanding advantages, including large surface areas, high reactivity, high surface acidity, non-toxicity, easy availability, low cost, environmental friendliness, and high chemical and mechanical stability.¹⁹⁾ To the best of our knowledge, acid-activated clays have not yet been extensively studied for their potential to reduce Cs^+ ions in aquatic environments. In removing Cs^+ ions by acid-activated clay, it is estimated that removal efficiency depends on several factors such as the type of acid, the concentration of acid, the nature of clay, the treatment period, and temperature.

In this study, authors activated the natural kaolinite clay through acid treatment by sulfuric acid. Then, we explored the possibility of removing radioactive Cs^+ ions from aqueous solution using both natural kaolinite clay and the resulting acid-activated kaolinite clay and compared their adsorption capacity. The influence of various operating parameters including the contact time, dose of adsorbent, and initial Cs^+ ions concentration was completely investigated. Besides, adsorption behaviors of Cs^+ ions by natural kaolinite clay and the resulting acid-activated kaolinite clay were also interpreted in detail using adsorption isotherm and kinetic models.

2. Materials and Methods

2.1. Reagents

A cesium chloride salt (CsCl), analytical grade reagent, was supplied by Daejeong Chemicals & Metals, Co, Ltd. Siheung-si, Korea. A stock solution containing 1000 mg L^{-1} of Cs^+ ions was made by diluting above reagent with distilled water. The concentrations of Cs^+ ions required for this study were prepared from a stock solution through successive dilutions. Reagent grade H_2SO_4 (95-97%, Merck, Darmstadt, Germany) was used for acid treatment of natural kaolinite clay.

2.2. Materials

The natural clay obtained from the coastal area of Pohang in Korea was a kaolinite ($\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$) type. The 500 g of natural kaolinite clay was crushed and pulverized into a powdery form. Subsequently, the natural kaolinite clay was washed thoroughly with distilled water to remove the impurities and then dried in an oven at 105°C for 1 day until a constant weight is attained. The dried natural kaolinite clay was stored in an air-tight glass jar and used as a sample for

the H₂SO₄ treatment.

2.3. Acid activation of natural kaolinite clay

The 50 g of natural kaolinite clay was added into a 500 mL of H₂SO₄ (2 M) in a round-bottom flask. The solution was shaken vigorously using a mechanical stirrer while heating at 80°C for 6 h at atmospheric pressure. The acidified clay was washed several times with deionized water for the purpose of eliminating the remaining sulfate ions and then dried in an oven for 2 h at 105°C. The natural kaolinite clay and acid-activated kaolinite clay were hereafter labelled KC and AKC, respectively. The KC and AKC were sieved to a fine particle size in the range of 100 μm to 200 μm using a mechanical sieve shaker. The KC and AKC obtained from sieving were employed as adsorbents for the adsorption experiments of Cs⁺ ions in aqueous solution.

2.4. Analytical methods

The chemical compositions were accomplished using X-ray fluorescence spectroscopy (XRF, ZSX Primus II, Rigaku, Japan). The BET surface area and pore size distribution were determined with N₂ adsorption isotherm measured by a specific surface area analyzer (BET, 3 Flex, Micrometrics, Atlanta, USA). Elemental analyses were performed by energy dispersive X-ray spectroscopy (EDX, X-MaxN, Oxford, UK). The pH value was measured using a pH meter (Radiometer, PHM 250 ion analyzer, Woonsocket, USA). Cation exchange capacity (CEC) were quantified by ion chromatography with a conductivity detector (940 professional IC Vario, Metrohm, Swiss). The concentration of Cs⁺ ions was analyzed by an inductively coupled plasma/atomic emission spectrometer (ICP/AES, Flame Modula S, Spectro, Kleve, Germany).

2.5. Adsorption experiments

Adsorption of Cs⁺ ions onto KC and AKC were performed in batch system at room temperature. In this adsorption experiment, the effects of the contact time, the initial concentration of Cs⁺ ions, and the dose of the adsorbent were evaluated. In addition to this, adsorption process was also analyzed by using various isotherms and kinetics. The adsorption was carried out in a flask by adding different amounts of adsorbents (0.5 g~5.0 g) to 500 mL of Cs⁺ ions solution with initial concentration (25 mg L⁻¹~100 mg L⁻¹) without controlling pH. The flask was shaken at 200 rpm using a Jar tester (Lab Tech, Korea) for a fixed time interval. A

fraction of solution was collected at each given contact time and then centrifuged. After that, the Cs⁺ ions remaining in the supernatant were determined using an ICP/AES. All adsorption experiments were performed in triplicate. Adsorption efficiency (η) was expressed as a percentage of adsorbed Cs⁺ ions compared to initial Cs⁺ ions, as shown in Eq. (1).

$$\eta(\%) = \left(\frac{c_0 - c_e}{c_0} \right) \times 100\% \quad (1)$$

where, C_0 (mg/L) and C_e (mg/L) are the initial and residual concentrations of Cs⁺ ions, respectively.

2.6. Adsorption isotherms

Two widely used Freundlich and Langmuir isotherms models were applied to our experimental data to obtain more behaviors associated with Cs⁺ ion adsorption on KC and AKC. These isotherm models illustrate the relationship between the amount of adsorbate adsorbed per adsorbent weight and the affinity of adsorbent and adsorbate when the adsorption reaches equilibrium at a certain temperature.¹⁹⁾ The isotherm model suitable for describing the experimental data depends on the correlation coefficient (R^2) of the profile that represents the linear regression obtained from the data. The Freundlich isotherm model is an empirical formula assuming that adsorption of adsorbate occurs on the surface of adsorbents with very heterogeneous properties.

The Freundlich isotherm equation is defined as $q_e = K_F \cdot C_e^{1/n}$, and by taking logarithms on both sides, it can be converted into a linear equation as shown in Eq. (2).

$$\log q_e = \log K_F + \frac{1}{n} \log C_e \quad (2)$$

where, K_F (mg/g) is a Freundlich adsorption constant representing the adsorption capacity of the adsorbent, and the larger the constant value, the better the adsorption capacity. n represents the magnitude of the adsorption affinity. q_e (mg/g) is the amount of adsorbate adsorbed per unit mass of adsorbent at equilibrium, which is expressed as Eq. (3).

$$q_e = \frac{(c_0 - c_e) V}{M} \quad (3)$$

where, V (L) is the volume of the solution and M (g) is the amount of adsorbent.

The Langmuir adsorption isotherm, expressed as Eq. (4), assumes that the adsorption is formed as a monolayer without

any attraction action between the adsorbed adjacent adsorbates.

$$\frac{C_e}{q_e} = \frac{1}{QK_L} + \frac{C_e}{q} \quad (4)$$

where, Q (mg/g) is the maximum adsorption amount of the adsorbent and K_L is the Langmuir constant representing the affinity of the adsorbate for the adsorbent.

Langmuir constant K_L is related with Gibbs free energy at the standard state, as shown in Eq. (5). Where, R is the ideal gas constant (8.314 J/mol·K) and T is the temperature (K). Generally, the range of Gibbs free energy for physisorption is between -20 and 0 kJ/mol, while chemisorption is between -80 and -400 kJ/mol.

$$\Delta G = -RT \ln(1000K_L) \quad (5)$$

2.7. Adsorption Kinetics

Adsorption kinetic study provides various information such as the performance of the adsorbents, the adsorption rate, and the mass transfer mechanism.²⁰⁾ In this study, the well-known pseudo-first-order and pseudo-second-order kinetic models were selected to fit the adsorption experimental data.

The equation of pseudo-first-order kinetic model is expressed as Eq. (6)

$$\frac{dq_t}{dt} = k_1(q_e - q_t) \quad (6)$$

Eq. (6) can be transformed as Eq. (7), which is a linear equation of logarithmic function.

$$\log(q_e - q_t) = \log q_e - \frac{k_1 t}{2.303} \quad (7)$$

where, q_t (mg g⁻¹) is the amounts of adsorbate adsorbed on per gram of adsorbent at time t (min). k_1 (min⁻¹) is the rate

constant of the pseudo-first-order kinetic. The calculated equilibrium adsorption amount ($q_{e,cal}$), the rate constant of pseudo-first-order kinetic model (k_1) and correlation coefficient (R^2) are established from the straight line of the relationship between $\log(q_e - q_t)$ and t .

The equation of pseudo-second-order kinetic model is given as Eq. (8),

$$\frac{dq_t}{dt} = k_2(q_e - q_t)^2 \quad (8)$$

Eq. (8) can be converted into a linear equation as Eq. (9).

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{1}{q_e} t \quad (9)$$

where, k_2 (g mg⁻¹ min⁻¹) is the rate constant of pseudo-second-order kinetic. The calculated equilibrium adsorption amount ($q_{e,cal}$) and the second order constant (k^2) and correlation coefficient (R^2) can be determined from the linear plot of t/q_t versus t .

3. Results and Discussion

3.1. Mechanism of sulfuric acid activation

Figure 1 shows the structural changes of KC by the attack of H₂SO₄. The clay mentioned above is a mineral with a hexagonal plate-like structure consisting of three layers of a silicon tetrahedral, an aluminum octahedral, and a silicon tetrahedral.²¹⁾ During H₂SO₄ treatment of KC, naturally occurring exchangeable interlayer cations such as Na⁺, K⁺, and Ca²⁺ ions are replaced by H⁺ ions attacking the layers, and, at the same time, creates the partial dissolution of Al₂O₃ as well as CaO and MgO from the octahedral layers.^{22,23)} This phenomenon causes the octahedral layers to open, resulting in an increase in internal surface area. The chemical reaction

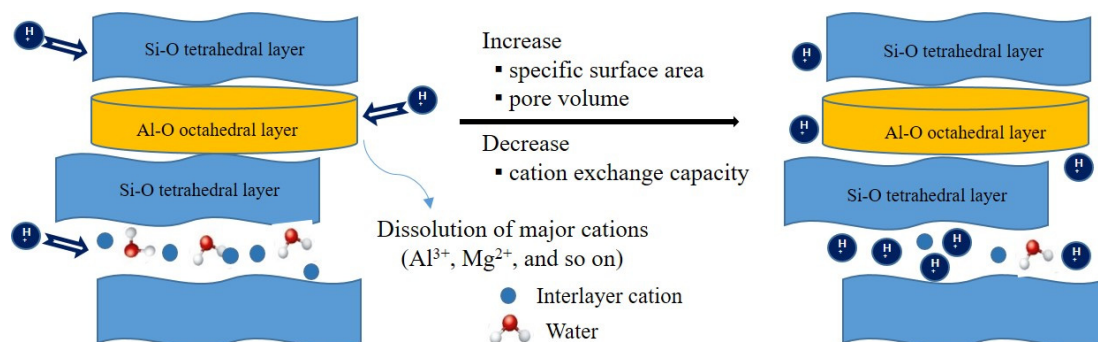
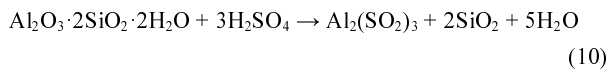


Fig. 1. Structural changes of natural kaolinite clay (KC) by the attack of H₂SO₄.

between KC and H₂SO₄ can be shown as Eq. (10).



Thereafter, Ca²⁺ and Mg²⁺ ions located on the surface of the layers are gradually exchanged with hydrogen ions from H₂SO₄. As a result, the AKC reaches almost saturation with H⁺ ions and, for that reason, it exhibits the acidic and better adsorption properties.

3.2. Characterization

Chemical compositions, CEC, and physicochemical properties of KC and AKC were shown in **Table 1**. The XRF was used to identify oxide compositions of KC and AKC. The major oxides in adsorbents were Al₂O₃, SiO₂, SO₃, and Fe₂O₃, along with minor oxides. As seen in **Table 1**, it was observed that the weight percentage of all the oxides in the AKC except

SiO₂ and TiO₂ was reduced as compared with those of KC. From this result, it is clearly shown that the leaching of cations from the layers causes the opening of the layer and the increase in the inner surface area by modifying the structure of the KC. In addition, it was seen that the SiO₂ content of AKC increased from 30.80 wt.% to 72.97 wt.%, whereas the Al₂O₃ content decreased from 19.44 wt.% to 12.33 wt.%. This is due to the partial dissolution of the Al-O octahedral layer and the exposure of Si-O tetrahedral layer in H₂SO₄ solution. As a result, it was found that AKC had a higher SiO₂/Al₂O₃ ratio rather than KC, showing an approximately fourfold increase.

According to CEC analysis, there was a significant change in CEC when natural kaolinite clay was treated with H₂SO₄. The CECs of KC and AKC were measured as 283.89 and 120.79 meq/100 g, respectively. The CEC reduction in AKC is likely due to the replacement of the major cations (Na⁺, Ca²⁺, K⁺, Mg²⁺) present between the layers of the KC with protons from H₂SO₄. In addition, the presence of excess SiO₂ can also reduce the CEC of AKC. Considering the above, it is inferred that the AKC is more acidic.

The BET surface area and total pore volume of AKC were measured as 252.69 m²g⁻¹ and 0.28 cm³g⁻¹, respectively. Their values were about 3 times higher in comparison with those of the KC. On the other hand, the pore size of AKC was slightly reduced. All of these changes are related to the dissolution of the impurities and the leaching of metal ions from octahedral layers during H₂SO₄ activation, which could develop not only more porous structure and surface area but also smaller particles. The pH value of AKC was 2.9, indicating it can be regarded as an acidic adsorbent. The low pH on AKC might be due to the deposition of a large amount of acidic functional groups on the surface during oxidation. As a consequence, surface acidity increased on AKC and that led to increase the anionic properties. This characteristic of AKC could improve the anion exchange capability with cesium cation during the adsorption process.

The elemental components of KC and AKC were determined by EDX and their analysis results were listed in **Table 2**. The

Table 1. Chemical compositions, CEC, and physicochemical properties of natural kaolinite clay and acid-activated kaolinite clay.

Items	Natural Kaolinite clay	Acid-activated kaolinite clay	
Major oxides (wt. %)	Na ₂ O	2.97	1.08
	MgO	1.82	0.66
	Al ₂ O ₃	19.44	12.34
	SiO ₂	30.80	72.97
	SO ₃	13.38	2.49
	K ₂ O	1.83	1.26
	CaO	4.65	2.83
	TiO ₂	1.36	2.26
	MnO	0.47	0.02
	Fe ₂ O ₃	21.98	3.44
SiO ₂ /Al ₂ O ₃	1.58	5.91	
Cation exchange capacity (meq/100 g)	283.89	120.79	
BET surface area (m ² g ⁻¹)	88.60	252.69	
Pore volume (cm ³ g ⁻¹)	0.10	0.28	
Pore size (Å)	86.29	62.61	
pH	6.4	2.9	

Table 2. Elemental component analysis of natural kaolinite clay and acid-activated kaolinite clay by EDX.

Adsorbent	O	Na	Mg	Al	Si	S	K	Ca	Ti	Fe
	Weight (%)									
Natural Kaolinite clay	54.48	2.54	1.22	9.23	11.37	4.60	0.94	2.37	0.76	11.50
Acid-activated Kaolinite clay	58.90	1.25	-	5.45	28.90	0.8	0.81	1.33	0.88	1.87

main elements constituting KC and AKC were O, Al, Si, S, and Fe. The weight percentage of Al, Ca, Mg, and Fe in AKC was observed as lower than those of KC. Whereas, Si showed the opposite trend due to its low solubility in H_2SO_4 solution. The data obtained are in close agreement with the results by XRF analysis. When KC is treated with H_2SO_4 , above-mentioned elements are more easily eluted, causing deformation of the layers of KC. This phenomenon results in the further formation of pore as well as the increase in the surface area. From the discussion above, it was revealed that AKC could be a better adsorbent than KC.

3.3. The removal mechanism of Cs^+ ions

The Cs^+ ions are more likely to form a chemical, more likely forming covalent bond at the apices of the tetrahedral $(\text{SiO}_4)^{4-}$ or the octahedral $[\text{AlO}_3(\text{OH})_3]^{6-}$ sheets available with the oxygen or hydroxyl groups, which are completely dissociated and form a stable adsorbing phase. The possible mechanism could be described by Eqs.11 and 12:



where, Y denotes the Al and Si present for the tetrahedral and octahedral sheets, respectively.

3.4. Effect of adsorbent dose

The effect of the adsorbent dose on the adsorption of Cs^+ ions in a 50 mg L^{-1} of Cs^+ ions solution was studied by

changing the amount of adsorbent from 0.5 g to 5.0 g. **Fig. 2** shows the concentration of Cs^+ ions remaining in the solution at a given contact time (min). As shown in **Figure 2**, the concentration of Cs^+ ions in the solution decreased rapidly within 1 minute of contact time and reached equilibrium in about 60 minutes regardless of the dose of the added adsorbent. **Figure 3** shows the adsorption efficiencies of Cs^+ ions according to the change in adsorbent dose at 60 min of contact time. As mentioned earlier, during activation of the KC using H_2SO_4 , the exchangeable interlayer cations are replaced with protons and a part of the octahedral cations are dissolved, resulting in new acid sites in the structure. This makes the KC more porous and acidic. Due to this reason, acid activation increases the number of sites responsible for adsorption of Cs^+ ions. At the dose of 1 g, the AKC removed

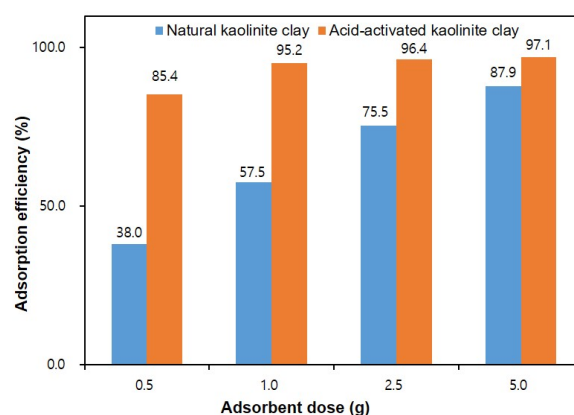


Fig. 3. Adsorption efficiency of Cs^+ ion according to change in dose of natural kaolinite clay and acid-activated kaolinite clay at equilibrium.

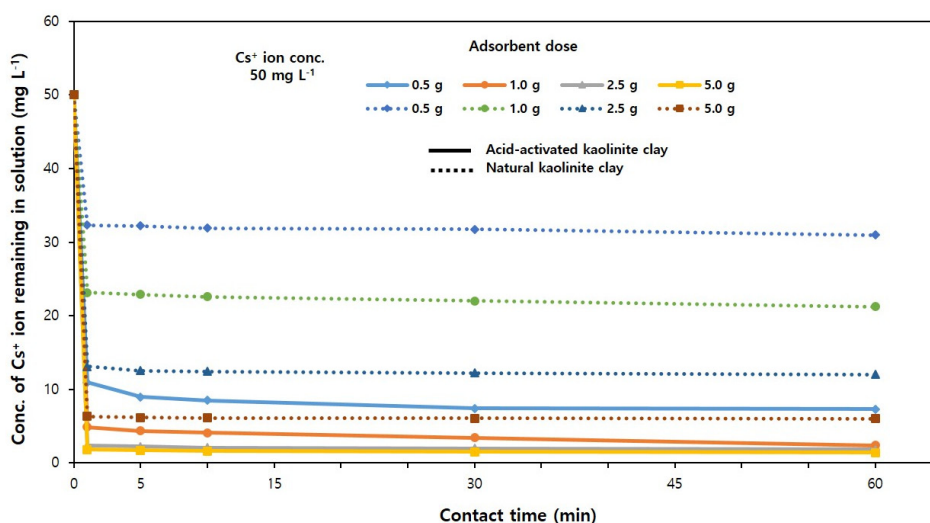


Fig. 2. Concentration of Cs^+ ions remaining in solution according to change in adsorbent dose at a given contact time (min).

85.4% of Cs^+ ions, while the KC removed 38.0%. By comparing the concentration of adsorbed Cs^+ ions per unit mass, it was found that the AKC had a higher adsorption capacity than the KC. Furthermore, the adsorption efficiency of Cs^+ ions by AKC tended to increase with increasing the dose, but the adsorption efficiency did not increase proportionally. For example, the adsorption efficiencies of AKC for Cs^+ ions were 95.2% and 97.1% at the dose of 1 g and 5 g, respectively. These experimental results showed that the higher adsorbent dose inhibited the increase in the adsorption ratio of Cs^+ ions. It is presumed that the decrease in the available adsorption site of AKC is caused by the aggregation of AKC at higher doses. Therefore, when removing Cs^+ ions using AKC, the appropriate dose for a given concentration of Cs^+ ions seem to be more effective from an economic point of view.

3.5. Effect of initial Cs^+ ions concentration

The effect of various initial Cs^+ ion concentrations in the range of 25 mg L⁻¹ to 100 mg L⁻¹ on 1 g of the adsorbent dose at the designated contact time (min) was studied, and the results are represented in **Figure 4**. As can be seen, the initial Cs^+ ion concentration in the solution decreased rapidly at the 1 min of the contact time, and then gradually decreased as the contact time elapsed and reached the equilibrium at about 60 minutes. **Figure 5** shows the adsorption efficiency of Cs^+ ions by KC and AKC for different initial Cs^+ ion concentrations at a contact time of 60 min. As shown, the AKC showed the adsorption capacity of Cs^+ ions approximately 1.7 times higher than that of the KC at 25 mg L⁻¹ of initial concentration. The adsorption efficiencies of Cs^+ ions by AKC were 96.9%, 95.2%,

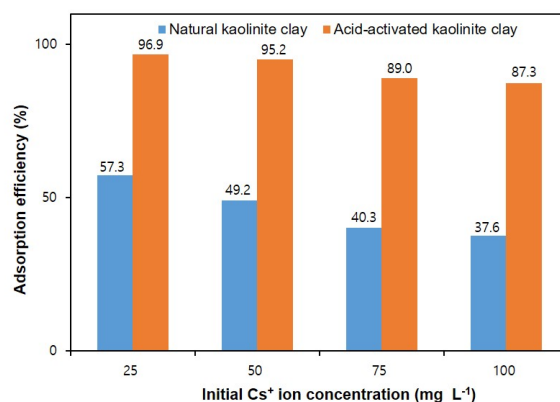


Fig. 5. Adsorption efficiency of Cs^+ ions by natural kaolinite clay and acid-activated kaolinite clay according to change in initial Cs^+ ions concentration at equilibrium.

89.0%, and 87.3%, respectively, at the initial Cs^+ ion concentrations of 25, 50, 75, and 100 mg L⁻¹. It was observed that the adsorption efficiency slightly decreased as the initial Cs^+ ion concentration increased. This is presumed to be due to the gradual increase in the Cs^+ ion concentration gradually saturating the active adsorption site in the fixed amount of adsorbent. For this reason, it was found that even if the concentration of Cs^+ ions was further increased, it had no effect on the improvement of the adsorption capacity of the AKC.

3.6. Adsorption isotherms

The adsorption data of Cs^+ ions in aqueous solution by KC and AKC were applied to Freundlich and Langmuir isotherm models, respectively. In general, R^2 is a measure of the fitness of the regression linear equation, and a value close to 1 indicates that the isotherm model is more suitable for the

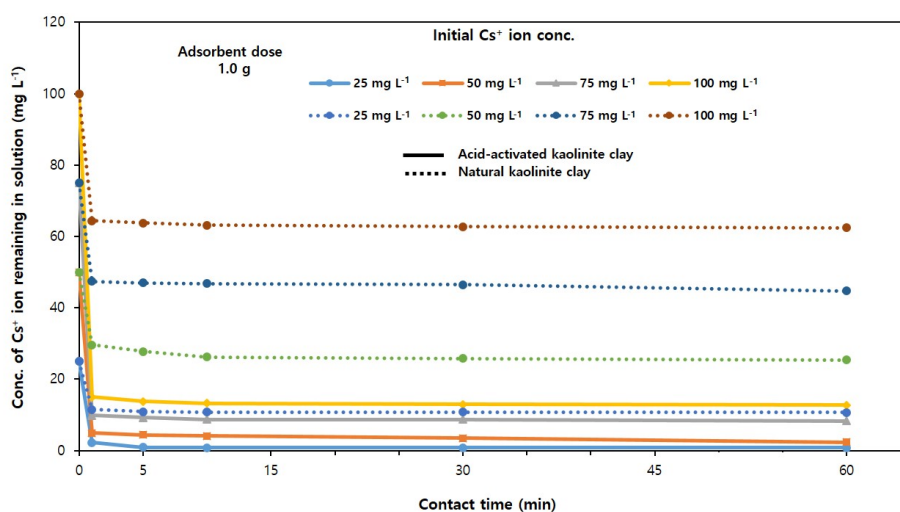


Fig. 4. Concentration of Cs^+ ions remaining in solution according to change in initial Cs^+ ion concentration on a fixed adsorbent dose at a given contact time (min).

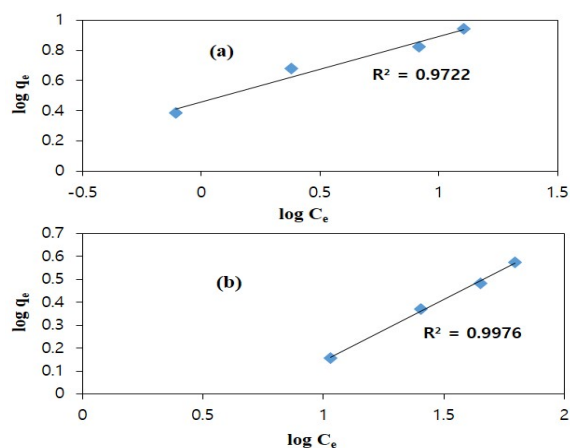


Fig. 6. Freundlich adsorption isotherm plots for the adsorption of Cs⁺ ions on (a) natural kaolinite clay and (b) acid-activated kaolinite clay.

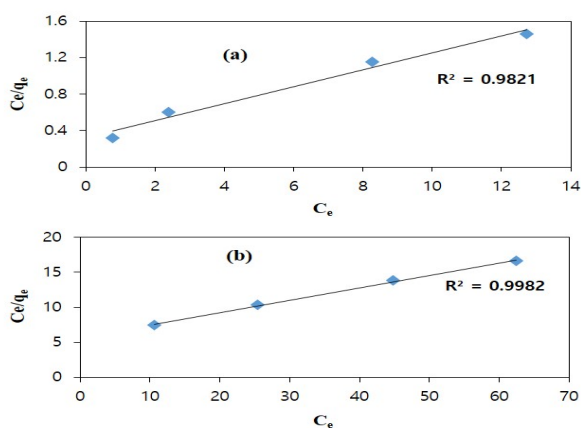


Fig. 7. Langmuir adsorption isotherm plots for the adsorption of Cs⁺ ions on (a) natural kaolinite clay and (b) acid-activated kaolinite clay.

experimental data obtained. The Freundlich and Langmuir adsorption isotherm plots for the adsorption of Cs⁺ ions on KC and AKC are shown in **Figures 6** and **7**, respectively. As shown, the R^2 values of the Freundlich and Langmuir adsorption isotherms were 0.9722 and 0.9821 for KC, and 0.9976 and 0.9982 for AKC, respectively. The R^2 values obtained from the Langmuir adsorption isotherm were slightly larger and close to 1.0 compared to Freundlich adsorption isotherm. From these results, it was found that the Langmuir isotherm model fits a little better with respect to the adsorption behavior of Cs⁺ ions.

The Freundlich constants K_F and $1/n$ and Langmuir constants Q and K_L values were calculated from equations 3 and 5, and the results were listed in **Table 3**. Freundlich and Langmuir constants, $1/n$ and K_L indicate the affinity of the adsorbate for adsorbent. In general, it is known that adsorption is effective when $1/n$ is in the range of 0.1 to 0.5, and adsorption is poor when it is 2 or more. According to this study, the Freundlich and Langmuir constants, $1/n$ and K_L values were found to be 0.25 and 0.46 for KC and 0.19 and 0.34 for AKC, respectively. Based on the above figures, it was clear that the Cs⁺ ions were favorably adsorbed by both KC and AKC. The Langmuir constants Q for KC and AKC were 5.65 mg/g and 10.6 mg/g, respectively, indicating that AKC can adsorb Cs⁺ ions approximately twice as much as KC. This suggests that AKC is more effective than KC when adsorbing Cs⁺ ions from aqueous solutions.

The ΔG values of KC and AKC obtained using Eq. (6) was -15.12 kJ/mol and -14.45 kJ/mol, respectively. The negative value of ΔG confirmed that the adsorption process of Cs⁺ ions by KC and AKC is spontaneous and proceeds by physical adsorption.

3.7. Adsorption Kinetics

The correlation coefficient (R^2) obtained from the straight plots of $\log(q_e - q_t)$ versus t , and t/q_t versus t was represented in **Figs. 8** and **9**, respectively and the calculated parameters of the kinetic models were listed in **Table 4**. As seen in **Figures 8** and **9**, and **Table 4**, it was found that the correlation coefficients of pseudo-second-order kinetic model are almost equal to unity and furthermore, the obtained values ($q_{e,cal}$) agrees well with ($q_{e,exp}$) compared with those of pseudo-first-order kinetic model. From these reasons, it is judged that the pseudo-second-order kinetic model is more applicable to the adsorption of Cs⁺ ions on KC and AKC over the entire range of processing time. The applicability of the pseudo-order kinetic model in the adsorption of Cs⁺ ions by the KC and AKC indicates that chemisorption is a predominant process involved with the valency forces through the sharing or exchange of electrons between the adsorbent and the Cs⁺ ions, which suggests that the adsorbing Cs⁺ species are adsorbed

Table 3. Freundlich and Langmuir adsorption isotherm constants.

Adsorbent	Adsorbate	Freundlich constants		Langmuir constants	
		K_F (mg/g)	$1/n$	Q (mg/g)	K_L
Natural kaolinite clay	Cs ⁺ ions	1.37	0.25	5.65	0.46
Acid-activated kaolinite clay		2.80	0.19	10.6	0.34

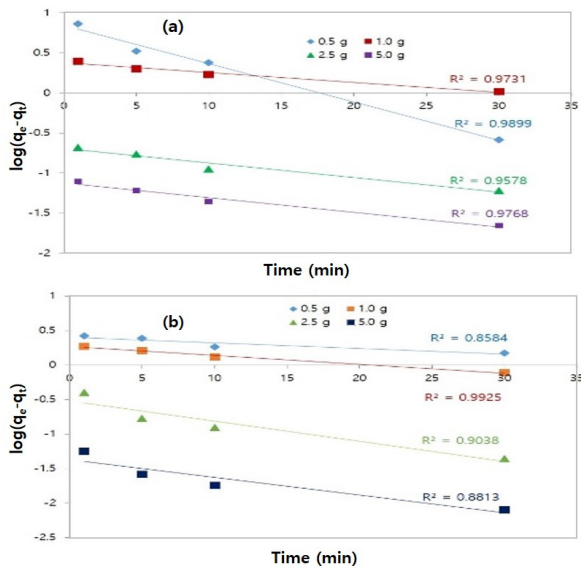


Fig. 8. Pseudo-first-order kinetic of Cs^+ ion adsorption according to change in dose of (a) natural kaolinite clay and (b) acid-activated kaolinite clay.

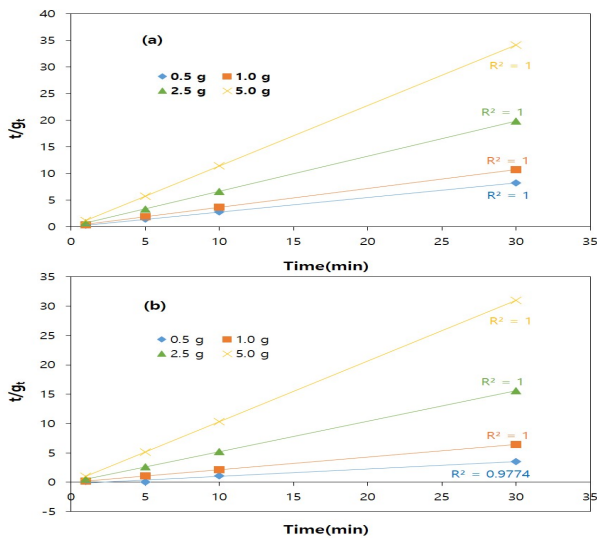


Fig. 9. Pseudo-second-order kinetic of Cs^+ ion adsorption according to change in dose of (a) natural kaolinite clay and (b) acid-activated kaolinite clay.

specifically onto the solid surfaces with strong chemical bonds. In addition, as the dose of AKC increased, the reaction rate value (k_2) also increased.

Furthermore, note that the AKC exhibited significantly enhanced uptake of Cs^+ ions compared to the precursor sample KC. However, the CEC values for the KC and the AKC were found to be 283.89 and 120.79 meq per 100 g, respectively. The CEC of the AKC were slightly less compared to the KC, but the sorption capacity of the AKC was increased significantly compared to the KC. The difference in CEC value could be explained by the facts that: the surface functional groups are greatly activated in the presence of H^+ ions and also some of interlayer cations are permanently replaced by the H^+ ions while conducting the acid activation, resulting in lower CEC value of the AKC. Moreover, the AKC possessed significantly higher specific surface area than the KC. The higher specific surface area greatly facilitated the uptake of Cs^+ ions by the AKC.

4. Conclusions

In this study, we compared the adsorption capacity for Cs^+ ions from aqueous solution using KC and AKC treated with H_2SO_4 (2 M) at 80°C for 6 h. While treating KC with acid, some cations such as Mg^{2+} , Al^{3+} , Mg^{2+} , Fe^{3+} , and Ca^{2+} ions from the crystal lattice of KC are partially eluted and substituted with H^+ ions, resulting in increase in the pore volume and the surface area through opening of crystal lattice. The results of analysis showed that the pore volume and the surface area of AKC were approximately three times higher compared with KC. The removal rate of Cs^+ ions from aqueous solution by KC and AKC was very fast in the first 1 minute of contact time, and then became slow and finally reached equilibrium in around 60 min. Generally, the

Table 4. Adsorption kinetic parameters for adsorption of Cs^+ ions on natural kaolinite clay and acid-activated kaolinite clay

Adsorbent	Adsorbate	Pseudo-first-order kinetic model				Pseudo-second-order kinetic model	
		Dose (g)	$q_{e, \text{exp}}$ (mg g^{-1})	$q_{e, \text{cal}}$ (mg g^{-1})	k_1 (min^{-1})	$q_{e, \text{cal}}$ (mg g^{-1})	k_2 ($\text{g mg}^{-1} \text{min}^{-1}$)
Acid-activated kaolinite clay	(50 mg L^{-1})	0.5	8.54	9.65	0.20	8.31	0.41
		1.0	4.76	10.7	0.06	4.57	1.75
		2.5	1.93	5.07	0.05	1.91	8.86
		5.0	0.97	1.90	0.03	0.97	22.60
Natural kaolinite clay	(50 mg L^{-1})	0.5	3.80	22.33	0.018	3.56	0.52
		1.0	2.87	8.87	0.037	2.72	0.72
		2.5	1.52	1.70	0.22	1.51	0.79
		5.0	0.88	1.03	0.19	0.88	1.05

adsorption efficiency of Cs^+ ions by AKC was relatively higher than KC in the experimental conditions conducted in this study. A detailed analysis of the correlation coefficient (R^2) revealed that Langmuir isotherm model adequately describe the adsorption data for adsorption of Cs^+ ions on both KC and AKC compared to Freundlich isotherm model. Based on the Langmuir isotherm constant Q , adsorption capacity of KC and AKC for Cs^+ ions was 5.65 mg g^{-1} and 10.6 mg g^{-1} , respectively. Kinetic studies represented that pseudo-second-order kinetic model was more suitable for adsorption behavior of Cs^+ ions on both KC and AKC than pseudo-first-order kinetic model due to not only the consistency of the experimental value ($q_{e, \text{exp}}$) and the calculated value ($q_{e, \text{cal}}$) but also higher correlation coefficient (R^2). This study has demonstrated that AKC could be effectively utilized for the removal of Cs^+ ions from aqueous solution rather than KC.

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Declaration of Competing Interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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